30Video Grayscale Installation Guide

Howdy! Thank you for purchasing my 30Video Grayscale card! – zigzagjoe :)

To install the card, follow these instructions carefully. **Please take care**: I am not responsible if you accidentally injure yourself or cause damage to your computer during the installation process. If you are at all uncomfortable with these procedures, please seek a qualified individual to assist you with the installation process.

In the Box

1. 30Video HC PDS card
2. IO plate
3. VGA cable
4. 2x M3 Philips screw and nuts

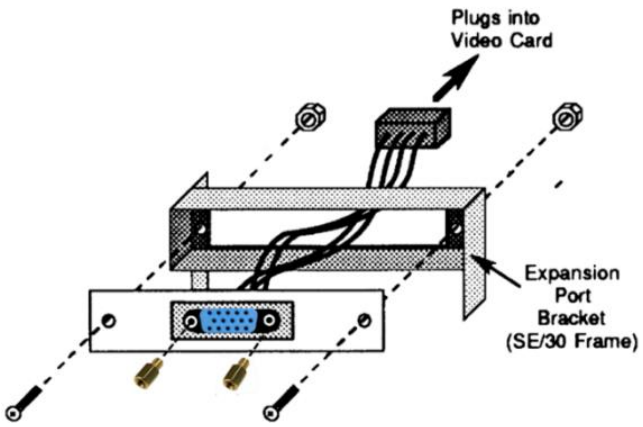
Optional Grayscale Kit

1. CRT Neckboard
2. Grayscale wire harness
3. Nonconductive adjustment tool

PDS Card Installation

Turn off and unplug your computer for at least 5 minutes before disassembling it.
Do not touch the analog board or CRT as it may remain charged after power is removed.

1. Remove plastic programmer's switch, if present.
2. Using a long Torx T-15 screwdriver, remove the 2 screws near the IO area and 2 screws under the handle, then remove the rear of the case.
3. Remove the 3 screws securing the expansion port bracket to the chassis.
4. Install the VGA cable into the IO cover, if needed
5. Attach the IO cover to your expansion port bracket using the provided screws.
6. Plug the VGA cable into the connector labeled "VGA" on the PDS card.
    a. ⚠ **Connector only fits one way**!
7. Seat the card **carefully** into the PDS slot on the motherboard.
    a. It may take some force, but it should seat smoothly.
    b. Ensure the logic board is entirely forward in the chassis rails if the card does not seem to be seating correctly. Check that no pins have become bent and nothing is stuck in your PDS slot.
8. **If you have the grayscale kit:** proceed to the grayscale kit installation instructions on following pages.
9. Attach the expansion port bracket to the chassis.
10. Remove the plastic IO port cover from the rear case by carefully pushing on it.
11. Reinstall the rear case, and tighten the 4 screws.



Stacking cards

The 30Video card may be "stacked" on other PDS cards with an appropriate PDS pass-through slot. When installing a second card, carefully bend the chassis card support tabs away from the cards so they do not make contact, and cover the tabs with insulating tape. 30Video supports slots ID **9** and **A**, selectable by DIP switch on the card.

Please note: Most systems should be able to support two cards, however, more than one card is not guaranteed to work and can cause system instability / inability to boot. A strong power supply is essential.

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Usage

No software is required to use the card – it's plug-and-play!

Connect a VGA cable and monitor to the VGA connector **prior to turning system power on**. **Do not plug VGA in while the system is powered.** Your monitor will be detected and turn on shortly after system boot.

If the grayscale harness is installed, the internal screen will have 256 shades of gray available while an external monitor is not connected. Some games work better in color modes; you may use any color mode as colors will be mapped to correct shades of gray. You can connect an external monitor *before powering the machine on*; the internal screen will revert to black and white while the external monitor is connected.

To change the active resolution, open Monitors control panel, select the 30Video monitor icon, and click the Options button. Pick the new resolution, then reboot your system. This will work on System 6.0.8+. Changing resolutions is not supported when internal grayscale is active.

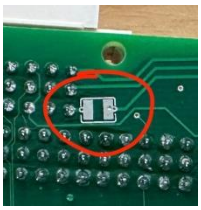
Gamma correction: VGA monitors will default to a linear gamma appropriate for **LCD** monitors. In grayscale mode, the internal screen will default to an appropriate correction table for the SE/30 CRT.

You may change the active gamma table by opening the Monitors control panel, hold the option key, then click the Options button. A linear (uncorrected) and CRT table are provided for both external VGA and internal grayscale modes. Additional gamma tables can be created using third party software.

Gamma correction is not available in Thousands of colors/grays modes. The internal CRT will be slightly dimmer.

Other Notes

- While 256 shades of grays are supported on external VGA monitors, only 16 shades of gray will be available. This is normal, and there is not a workaround. This limitation does not apply to internal grayscale.
- While using netBSD or MacsBug with internal grayscale, you may notice that the right margin of the display has turned on and has white or random contents. While strange looking, this is not harmful.
- **Users of Bolle's ethernet riser:** You **must** solder the 12V jumper on the rear of the riser, or you will not get video output from the 30Video card. See picture to right.



Compatibility notes

No compatibility issues are known at the time of writing this document.

- Supports System 6 - 8.1, A/UX (3.0, maybe others), netBSD
- Compatible with accelerators: tested with Carrera 040, Interware Booster 30-SE50F, DiimoCache 030
- Compatible with IIsi, SE/30, and third party ROMs in 24 bit and 32 bit modes.
- **Highly** recommended to have recapped analog & logic boards and recapped / retrofitted power supply.

Reporting issues

I've done the best I can to test this card thoroughly, but bugs are always possible. Please report any issues through this github repository. Updated firmware and flashing utilities will be posted here as well, and please feel free to ask questions.

https://github.com/ZigZagJoe/30Video-Support

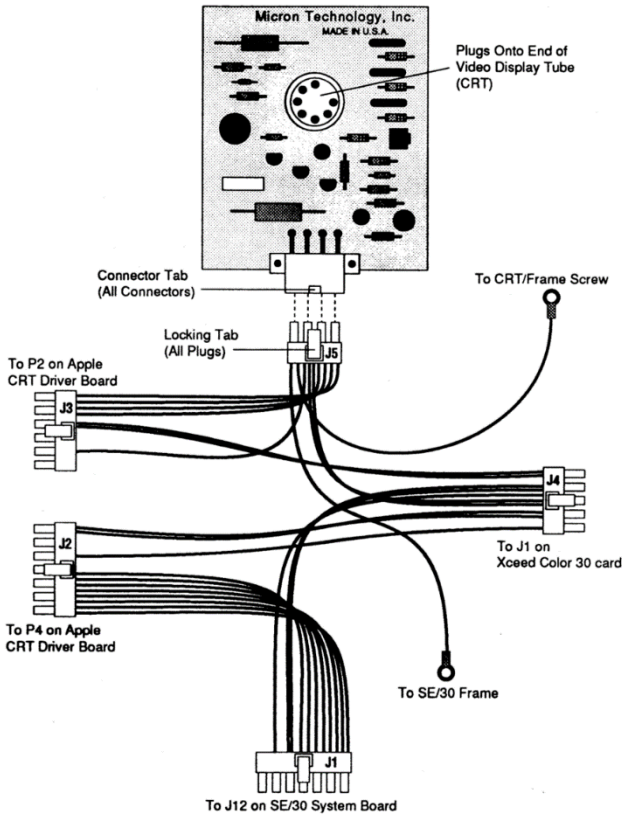
Happy Macing!! :)

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Grayscale Kit Installation

CAUTION: The CRT should be discharged before proceeding as it can retain charge while off!
The CRT neck is very fragile. USE EXTREME CAUTION while working around the CRT neck.
All connectors have a latch on them. Release the latch before pulling on the connector.

1. Disassemble the machine and install the PDS card as directed above.
2. Remove the upper-left torx screw securing the CRT to the frame to free the neck board's grounding lug.
3. With utmost caution, pull the apple neckboard straight off the back of the CRT tube
   DO NOT ROCK, WIGGLE, OR TWIST THE BOARD!! YOU *WILL* BREAK THE CRT!!
4. Unplug the connector from P4 on the analog board. The neckboard can now be removed from the machine.
5. Unplug the logic board cable from the logic board and unplug it from the analog board, then remove it.
6. You are now ready to install the grayscale harness.
7. Connect harness **J2** to **P4** on analog board (where stock logic board cable plugged in)
8. Connect harness **J3** to **P2** on analog board (where stock neck board cable plugged in)
9. Connect harness **J1** to **J12** on logic board
10. Reinstall the torx screw securing the CRT to the frame with the longer of the grounding cables attached to **J5**.
11. Attach **J5** to the new neckboard.
12. **With care,** seat the new neckboard onto the CRT neck.
    a. **The new CRT connector may be stiff. Ensure all the pins have seated into the connector correctly then carefully press it straight on.**
    b. You may first need to remove a plastic isolation element from the CRT pins: it pulls straight off.
13. Attach **J4** to the 30Video card seated in the PDS slot.
14. Reinstall the IO bracket to the chassis, with the last grounding lug under one of the screws to the frame.
15. Adjust the cutoff below, then re-assemble your computer.



Grayscale Cutoff Adjustment

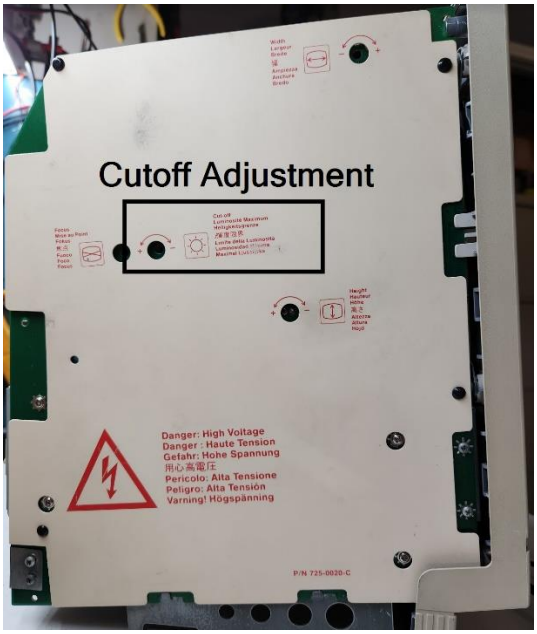
CAUTION: DO NOT TOUCH ANY METAL SURFACE INSIDE THE COMPUTER WHILE POWERED ON.

1. Power on your computer with the back removed.
2. Verify you have 256 shades of gray available in the Monitors control panel, but leave in B&W mode. Put something with high contrast on the screen. Allow machine to run for 5 minutes to warm up.
3. Adjust the external brightness control to maximum brightness.
4. **Using the provided nonconductive adjustment tool,** adjust the cut-off control located on the side of the analog board until brightness is to your preference. Stop before retrace lines (zigzag lines across screen) become visible, and black sections of the screen should remain dark. **See figures on following page.**
5. **Using the provided nonconductive adjustment tool,** adjust the potentiometer **RV1** on the right side of the neckboard to just before fine detail / text begins to "smear". Open other apps/windows & switch to 256 gray mode and verify no smearing appears, if so, reverse the potentiometer until it goes away.
6. The CRT image should not require further adjustments if it was in adjustment prior to installation. Please see Apple documentation if image size/position adjustments are required.

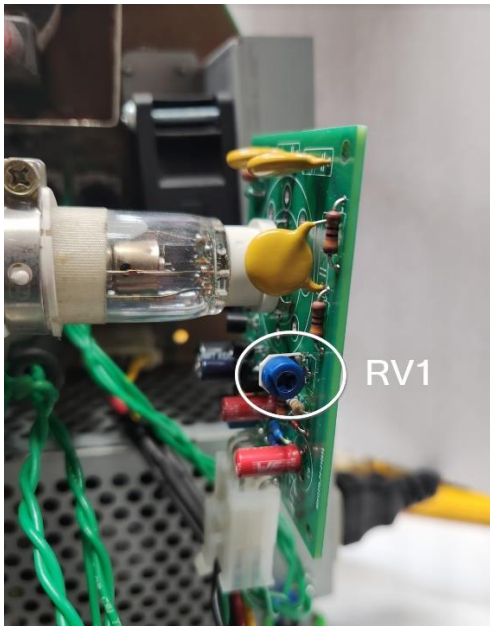
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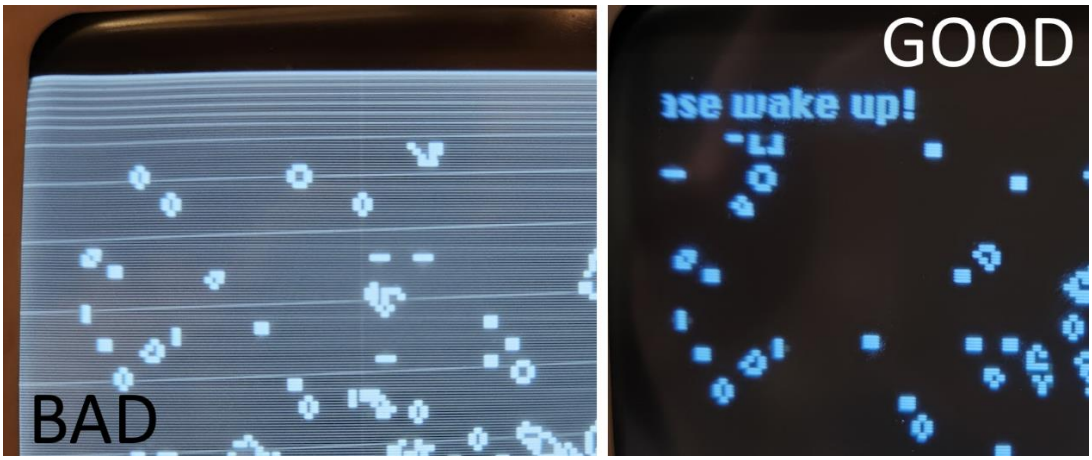
Main Cutoff Location



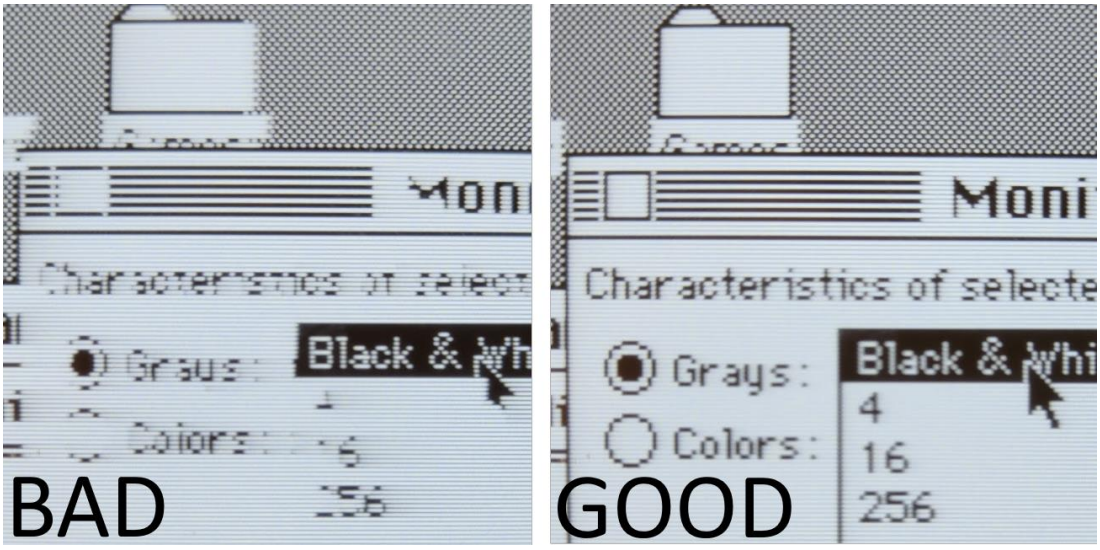
Neckboard adjustment location



Main Cutoff Adjustment



Neckboard adjustment



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